KOPAL KHARE

+91 9407091288 — kopalkhare2@gmail.com — linkedin.com/in/kopalkhare — github.com/kopalkhare2

SUMMARY

Computer Science undergraduate specializing in Artificial Intelligence & Machine Learning with strong foundations in Business Technology Solutions, enterprise data management, and Generative AI systems. AWS Certified Solutions Architect with practical expertise in LLM primitives , prompt engineering , autonomous agentic tool loops, and hybrid RAG architectures. Experienced in building end-to-end commercial solutions, automated data pipelines, and client reporting platforms across Python, SQL, Java (Spring Boot), Next.js, and PostgreSQL.

TECHNICAL SKILLS

GenAI & LLM Primitives: Transformers, Context Windows, Sampling , Prompt Engineering , Hallucination Mitigation, Prompt Injection Defense, PII Redaction
Agents, RAG & Frameworks: LangGraph, LangChain, RAG Pipelines , Vector Stores , Open-Source LLMs (Llama, Mistral, Gemma) via Inference APIs, RAGAS Evaluation Framework
Languages & Data Management: Python, SQL , Java, TypeScript, JavaScript
Backend, Cloud & Architecture: Spring Boot, FastAPI, Next.js , RESTful APIs, Microservices, AWS (EC2, S3), Docker, CI/CD (GitHub Actions)
Databases & Analytics: PostgreSQL, Redis, SQLite, Prisma ORM, Flyway Migrations, ACID Transactions, Reporting & BI Dashboards

PROJECTS

AK Saarthi AI – Autonomous Financial Advisory & BI Platform — Next.js 16, TypeScript, PostgreSQL, Gemini 1.5 Live Demo

- Delivered an end-to-end financial advisory solution providing commercial clients with interactive BI dashboards, multi-portfolio aggregations, policy tracking, and automated commission reporting.
- Engineered Gemini 1.5 Flash pipelines with few-shot prompting and structured JSON outputs for contextual financial summaries, insurance gap analyses, and automated renewal reminder drafting.
- Implemented edge HMAC-SHA256 JWT authentication, role-based client data isolation, 30-day automated renewal cron jobs, and validated API workflows with comprehensive Jest test suites.

PolicyLens – Multimodal Legal & Insurance Document RAG — Python, FastAPI, ChromaDB, LlamaIndex, RAGAS

- Architected an enterprise RAG service parsing 40+ page documents using layout-aware chunking, dense vector embeddings, and open-source LLMs (Mistral, Llama) via inference APIs.
- Mitigated hallucinations by applying Cross-Encoder reranking (bge-reranker) and exact clause coordinates; implemented Chain-of-Thought prompts enforcing strict schema validation.
- Established an automated evaluation harness with the RAGAS framework across 60 ground-truth test cases, achieving **0.93 Faithfulness** and **0.88 Context Precision**.

AgentForge – Autonomous Multi-Agent Workflow Engine — Python, FastAPI, LangGraph, Redis, Celery, Docker

- Designed stateful multi-agent execution graphs using LangGraph with autonomous tool-calling loops (Python REPL, database queries) to decompose multi-step business and analytical objectives.
- Constructed isolated Docker sandboxes with CPU/memory limits and human-in-the-loop approval checkpoints before executing destructive operations, ensuring enterprise data safety.
- Orchestrated background task dispatching with Redis and Celery, streaming intermediate reasoning steps in real time via Server-Sent Events (SSE).

ClearLedger – Financial Ledger & Reconciliation Platform — Java 17, Spring Boot, PostgreSQL, Docker, Flyway Live Demo

- Engineered a financial operations backend enforcing double-entry balance invariant checks, transactional transfers, and immutable append-only audit logs with zero data divergence.
- Built batch settlement ingestion and deterministic two-way reconciliation algorithms to automatically flag transaction discrepancies and streamline client dispute resolution.
- Implemented trial-balance validation, sub-50ms historical statements via indexed PostgreSQL schemas, and validated failure modes with **49/49 automated unit and integration tests** (JUnit5).

EXTRA-CURRICULAR ACTIVITIES

Blockchain Club June 2024 – Present
Core Member, VIT Bhopal University

- Co-organized 4+ national-level technical events with 500+ participants in collaboration with Accelchain and Notion.
- Coordinated event planning, technical execution, and cross-functional logistics with student teams and external collaborators.

British Airways Data Science Job Simulation May 2026
Forage Virtual Experience, Remote

- Analyzed customer review datasets using Python, performed exploratory data analysis, and built predictive models to identify actionable commercial business recommendations.

CERTIFICATIONS

AWS Certified Solutions Architect – Associate — IBM Machine Learning Specialization – Coursera — NPTEL Cloud Computing – IIT Kharagpur — NPTEL Internet of Things – IIT Kharagpur

EDUCATION

Vellore Institute of Technology July 2023 – July 2027
B.Tech in Computer Science and Engineering (Artificial Intelligence & Machine Learning) CGPA: 8.42/10